CS 5402: Introduction to Data Mining

SP2026

(Lecture -2)

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Data Types & Data Quality

- Review: Introduction to Data Mining

- Why understanding data matters before algorithms?

Example-1: Misinterpreting Attribute Meaning

ZIP Code Problem:

You are clustering customers using:

- ZIP Code
- Age
- Income

Algorithm-First Mistake

- Apply **k-means clustering** directly.

Why This Fails

- k-means computes **means**
- ZIP codes are **identifiers**, not quantities
- The “average ZIP code” is meaningless

Example 2: Ignoring Time in Time-Series Data

Situation

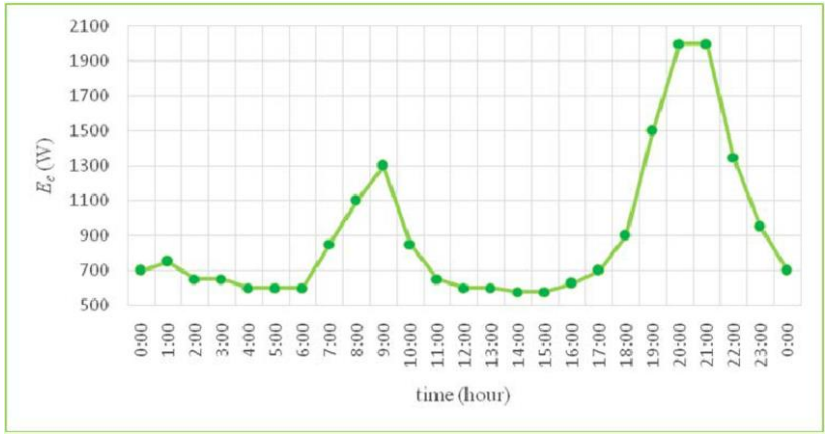
You have hourly energy consumption data.

Algorithm-First Mistake

- Treat each row as independent
- Apply standard classification/regression

Why This Fails

- Temporal order is ignored
- Trends and seasonality are lost



Example 3: Class Imbalance in Real-World Classification

Situation

Fraud detection dataset:

- 99.5% non-fraud
- 0.5% fraud

Algorithm-First Mistake

- Train a classifier
- Evaluate with accuracy

Why This Fails

- 99.5% accuracy by predicting “non-fraud”
- Model is useless

Why Data Matters in Data Mining?

- Data mining is **data-driven**, not algorithm-driven
- Poor data quality → misleading patterns
- Most real-world effort:
  - ~70–80% data understanding & preparation
- *Garbage in → garbage out*

Understanding data determines:

- What operations are meaningful
- Which algorithms are valid
- How models should be evaluated
- Whether results are trustworthy

Learning Objectives

By the end of this lecture, you will be able to:

- Identify common data types used in data mining
- Distinguish between attribute types
- Recognize data quality issues
- Apply basic data quality improvement techniques

Warm-Up Question

Which task typically takes the **most time** in a real-world data mining project?

A. Choosing the best algorithm

B. Training the model

C. Data understanding and preprocessing

D. Visualizing results



What is Data?

At its core, data is a collection of facts, figures, or pieces of information. In data science, we view it more formally.

- A dataset is a collection of ***data objects***, which are simply entities.
  - *Examples*: A customer, a patient, a house, a tweet.
- Each object is described by a set of **attributes**, also known as variables, features, or fields.
  - *Examples*: eye color of a person, temperature, name, student_id, etc.

Attributes: Describing the Objects

An ***attribute*** is a property or characteristic of a data object:

- Attribute is also known as variable, field, characteristic, dimension, or feature
- Consider a customer (the data object), what are their attributes?
  - *Examples*: address, phone number, gender, age, transaction history, etc.

A collection of attributes describe an ***object***

- Object is also known as record, point, case, sample, entity, or instance
- Attributes are often partial descriptions of an object, corresponding to the aspects that we can observe.

Attributes

Objects

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Quick Question: Object vs. Attribute

In a **movie recommendation dataset**, what is most commonly treated as the **object**?

A. Movie genre

B. User

C. Rating value

D. Timestamp

Types of Attributes

Attribute values are numbers or symbols assigned to an attribute for a particular object

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No

- We can split attributes into two main families:
  - Qualitative (Categorical): Describes a quality or characteristic. It represents categories or label.
    - *Example*: major (e.g., 'Computer Science'), eye_color (e.g., 'Blue')
  - Quantitative (Numeric): Describe a measurable quantity. It represents numbers.
    - *Example*: gpa (e.g., 3.8), height_in_cm (e.g., 175)

Why it matters:
Determines distance metrics
Determines applicable algorithms

Categorical Attributes

- Nominal: Categories with no intrinsic order. You can't say one is "greater" than another.
  - *Examples*: ID numbers, eye color, zip codes, major

- Ordinal: Categories with a meaningful order or rank.
  - *Examples*: year (Freshman < Sophomore), grade (A> B >C), t-shirt_size { S < M < L)

Numeric Attributes

- Interval
  - Differences meaningful, no true zero
  - Example: temperature (°C)
- Ratio
  - True zero exists
  - Examples: height, weight, income

- Discrete: Values are countable and can only take on specific, distinct numbers (often integers)
  - *Examples*: number_of_courses_taken (e.g., 3), age_in_years (e.g., 23). Calendar_date (e.g., 08/27/2025).
- Continuous: Values can take on any real number within a given range.
  - *Examples*: temperature_in_Celsius (e.g., 36.7), length_in_cm (e.g., 23.5)

Nominal levels of measurement

Values are distinct symbols

- Values themselves serve only as labels or names
- *Nominal* comes from the Latin word for name

Example: attribute "outlook" from weather data

- Values: "sunny","overcast", and "rainy"

Outlook	Temperature	Humidity	Windy	Play-time
Sunny	Hot	High	False	5
Sunny	Hot	High	True	0
Overcast	Hot	High	False	55
Rainy	Mild	Normal	False	40
...	...	...	...	...

- No relation is implied among nominal values (no ordering or distance measure)
- Only equality tests can be performed

Ordinal levels of measurement

- Impose order on values
- But no distance between values defined
  - Example: attribute "temperature" in weather data
    - Values: "hot" > "mild" > "cool"

Outlook	Temperature	Humidity	Windy	Play-time
Sunny	Hot	High	False	5
Sunny	Hot	High	True	0
Overcast	Hot	High	False	55
Rainy	Mild	Normal	False	40
…	…	…	…	…

- Note: addition and subtraction don't make sense
  - Example rule:

$$\text{temperature} < \text{hot} \Longrightarrow \text{play} = \text{yes}$$

Interval quantities

- Interval quantities are not only ordered but measured in fixed and equal units
  - Example: attribute "Temperature" expressed in degrees Fahrenheit
- Difference of two values makes sense
- Sum or product doesn't make sense

Outlook	Temperature (F)	Humidity	Windy	Play-time
Sunny	98	High	False	5
Sunny	90	High	True	0
Overcast	75	High	False	55
Rainy	70	Normal	False	40
…	…	…	…	…

Example:

- ❑ Today is 5 degrees hotter than yesterday. (YES)
- ❑ The combined temperature of yesterday and today is 188 F. (NO)

Ratio quantities

- Ratio quantities refer to measurements or values that express the relationship between two comparable quantities by division.
- They represent how much of one quantity exists relative to another and are often expressed as fractions, percentages, or proportions.

Example:

- "Price": 5 dollars per pound
- "Speed": 60 miles per hour
- "Density", 1 gram per cubic centimeter
- "Fuel Efficiency": 25 miles per gallon

- Ratio quantities are treated as real numbers
  - All mathematical operations are allowed
- Difference between Interval and Ratio:
  - A true zero point
  - The meaningfulness of ratios between values.

Properties of Attribute Values (1)

- The type of an attribute depends on which of the following properties/operations it possesses:
  - Distinctness: = ≠
  - Order: < >
  - Differences are meaningful : + -
  - Ratios are meaningful * /

- Nominal attribute: distinctness
- Ordinal attribute: distinctness & order
- Interval attribute: distinctness, order & meaningful differences
- Ratio attribute: all 4 properties/operations

Properties of Attribute Values (2)

	Attribute Type	Description	Examples	Operations
Categorical Qualitative	Nominal	Nominal attribute values only distinguish. (=, □)	zip codes, employee ID numbers, eye color, sex: {*male, female*}	mode, entropy, contingency correlation, $\chi 2$ test
	Ordinal	Ordinal attribute values also order objects. (<, >)	hardness of minerals, {*good, better, best*}, grades, street numbers	median, percentiles, rank correlation, run tests, sign tests
Numeric Quantitative	Interval	For interval attributes, differences between values are meaningful. (+, -)	calendar dates, temperature in Celsius or Fahrenheit	mean, standard deviation, Pearson's correlation, *t* and *F* tests
	Ratio	For ratio variables, both differences and ratios are meaningful. (*, /)	temperature in Kelvin, monetary quantities, counts, age, mass, length, current	geometric mean, harmonic mean, percent variation

Key Messages for Attribute Types

- The types of operations you choose should be "meaningful" for the type of data you have
  - Distinctness, order, meaningful intervals, and meaningful ratios are only four (among many possible) properties of data
  - The data type you see – often numbers or strings – may not capture all the properties
  - Analysis may depend on other properties of the data
    - Many statistical analyses depend only on the distribution
  - What is meaningful can be specific to domain

Quick Question: Attribute Types

Which attribute type best describes **"customer satisfaction level: low, medium, high"**?

A. Nominal

B. Ordinal

C. Interval

D. Ratio



Quick Question: Attribute

Given a student performance dataset, identify the attribute types:

Attribute	Type
Student ID	Nominal
GPA	Ratio (Continuous)
Major	Nominal
Year (Freshman–Senior)	Ordinal
Number of absences	Ratio (Discrete)



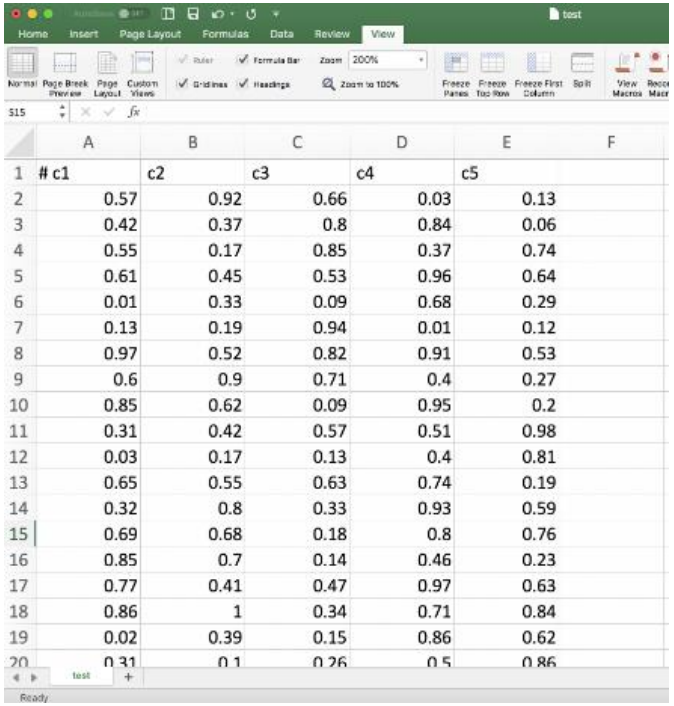
Types of Data by Domain

- Record data (tabular)
- Transaction data (shopping baskets)
- Text data (documents, emails)
- Time-series data (sensor signals, ECG)
- Graph data (social networks)
- Spatial data (GIS, maps)
- etc

Table

- Data that consists of a collection of records (rows), each with a fixed set of attributes (columns)
- This is the most common type of dataset.

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes



	A	B	C	D	E
1	# c1	c2	c3	c4	c5
2	0.57	0.92	0.66	0.03	0.13
3	0.42	0.37	0.8	0.84	0.06
4	0.55	0.17	0.85	0.37	0.74
5	0.61	0.45	0.53	0.96	0.64
6	0.01	0.33	0.09	0.68	0.29
7	0.13	0.19	0.94	0.01	0.12
8	0.97	0.52	0.82	0.91	0.53
9	0.6	0.9	0.71	0.4	0.27
10	0.85	0.62	0.09	0.95	0.2
11	0.31	0.42	0.57	0.51	0.98
12	0.03	0.17	0.13	0.4	0.81
13	0.65	0.55	0.63	0.74	0.19
14	0.32	0.8	0.33	0.93	0.59
15	0.69	0.68	0.18	0.8	0.76
16	0.85	0.7	0.14	0.46	0.23
17	0.77	0.41	0.47	0.97	0.63
18	0.86	1	0.34	0.71	0.84
19	0.02	0.39	0.15	0.86	0.62
20	0.31	0.1	0.26	0.5	0.86

Data Matrix

- If data objects have the same fixed set of numeric attributes, then the data objects can be thought of as points in a multi-dimensional space, where each dimension represents a distinct attribute
- Such a data set can be represented by an *m* by *n* matrix, where there are *m* rows, one for each object, and *n* columns, one for each attribute

	Sepal length X_1	Sepal width X_2	Petal length X_3	Petal width X_4	Class X_5
x_1	5.9	3.0	4.2	1.5	Iris-versicolor
x_2	6.9	3.1	4.9	1.5	Iris-versicolor
x_3	6.6	2.9	4.6	1.3	Iris-versicolor
x_4	4.6	3.2	1.4	0.2	Iris-setosa
x_5	6.0	2.2	4.0	1.0	Iris-versicolor
x_6	4.7	3.2	1.3	0.2	Iris-setosa
x_7	6.5	3.0	5.8	2.2	Iris-virginica
x_8	5.8	2.7	5.1	1.9	Iris-virginica
⋮	⋮	⋮	⋮	⋮	⋮
x_{149}	7.7	3.8	6.7	2.2	Iris-virginica
x_{150}	5.1	3.4	1.5	0.2	Iris-setosa



Transaction Data

- A special type of data, where
  - Each transaction involves a set of items.
  - For example, consider a grocery store. The set of products purchased by a customer during one shopping trip constitute a transaction, while the individual products that were purchased are the items.
  - Can represent transaction data as record data

TID	*Items*
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

⇒

X =

TID	Bread	Coke	Milk	...
1	1	1	1	...
2	1	0	0	...
3	0	1	1	...
4	1	0	1	...
5	0	1	1	...

Text Data

- Each document becomes a ‘term’ vector
  - Each term is a component (attribute) of the vector
  - The value of each component is the number of times the corresponding term occurs in the document.

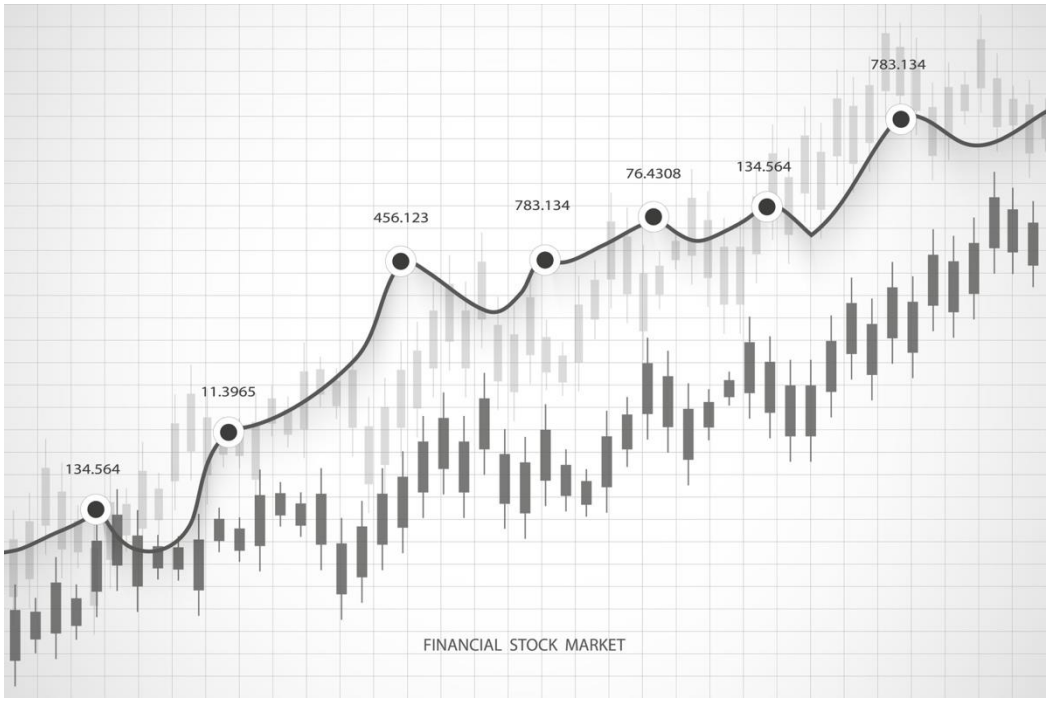


	team	coach	play	ball	score	game	win	lost	timeout	season
Document 1	3	0	5	0	2	6	0	2	0	2
Document 2	0	7	0	2	1	0	0	3	0	0
Document 3	0	1	0	0	1	2	2	0	3	0

Time Series Data

The order of the data objects matters.

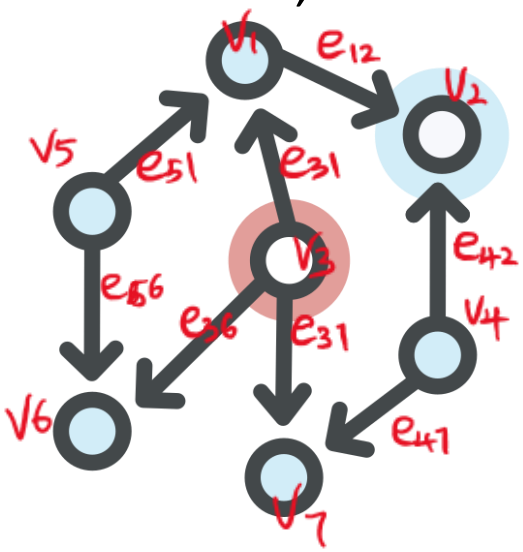
- **Temporal Data (Time Series):** A sequence of measurements taken over time.
- *Example:* Daily stock prices, temperature readings.



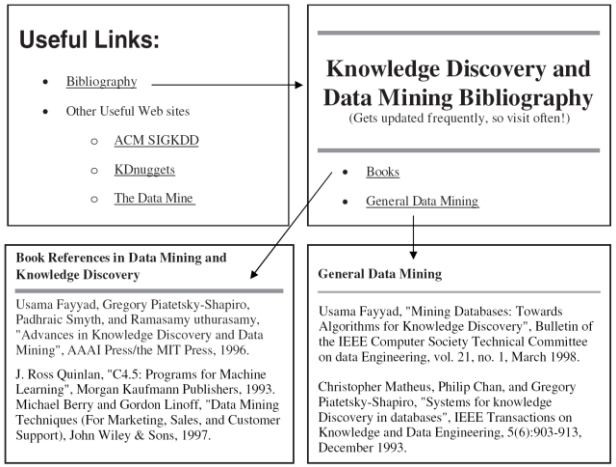
Graph Data

Used when the connections **between** objects are as important as the objects themselves.

- **Objects** are **nodes** (vertices).
- **Relationships** are **links** (edges).
- *Examples:* Social networks (you and your friends), molecular structures, road maps.



$G = \langle V, E \rangle$

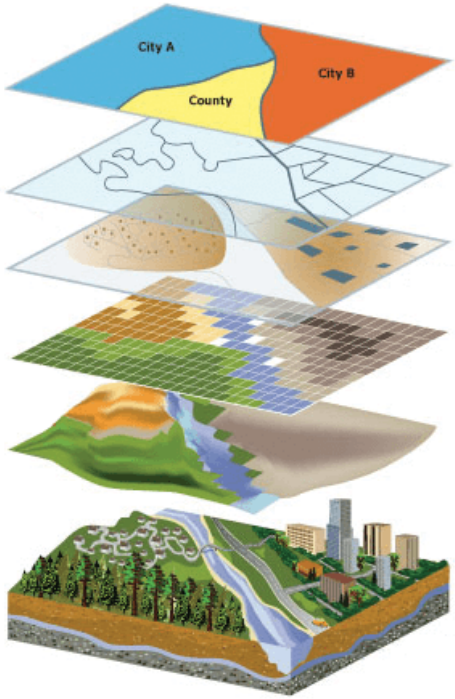




Spatial Data

The order of the data objects matters.

- Spatial Data: that has a geographic component.
- *Example:* GPS coordinates, weather maps.

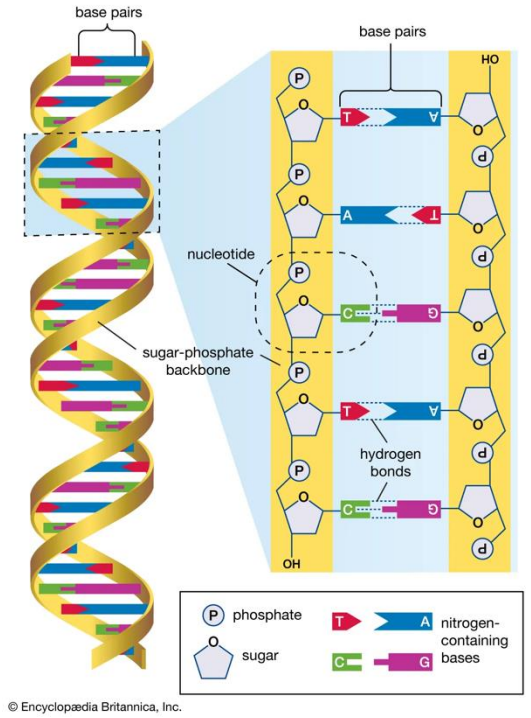


Sequential Data

The order of the data objects matters.

- Sequential Data: An ordered list of items.
- Example: Genomic sequence data

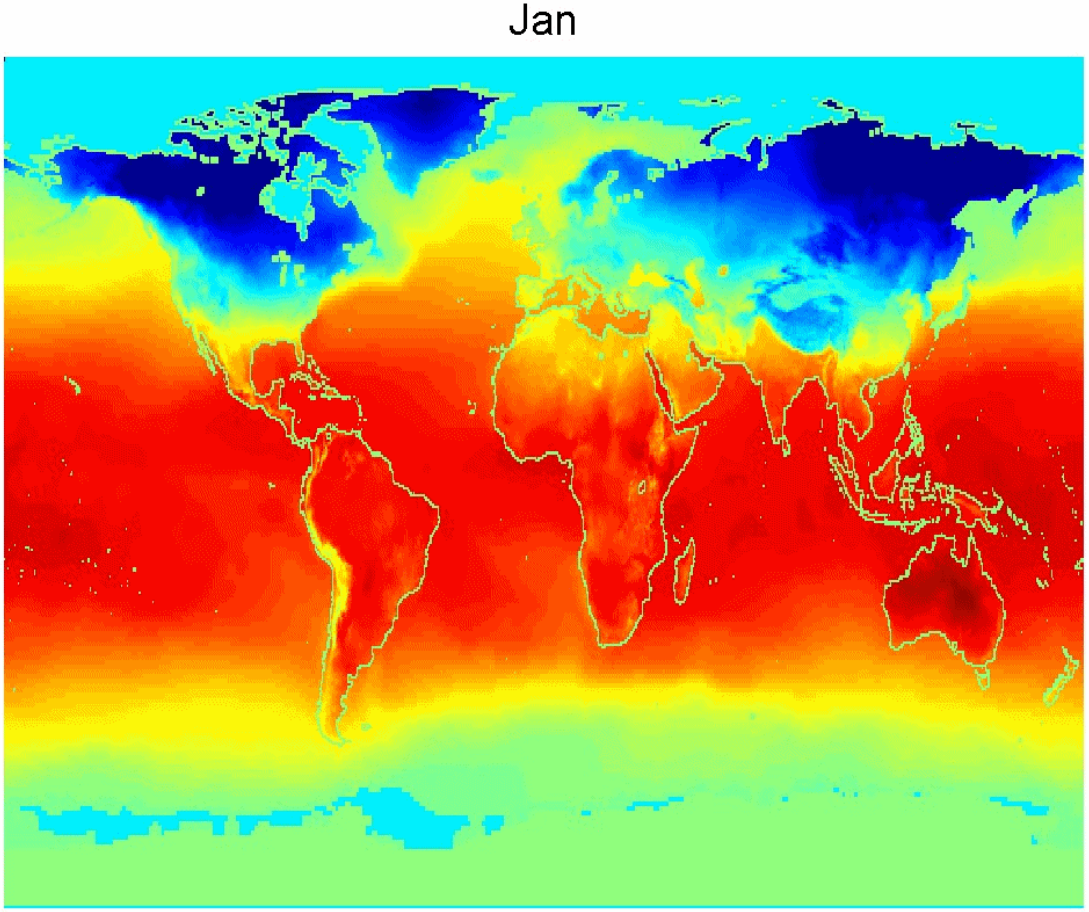
```
GGTTCCGCCTTCAGCCCCGCGCC
CGCAGGGCCCGCCCCGCGCCGTC
GAGAAGGGCCCGCCTGGCGGGCG
GGGGGAGGCGGGGCCGCCCGAGC
CCAACCGAGTCCGACCAGGTGCC
CCCTCTGCTCGGCCTAGACCTGA
GCTCATTAGGCGGCAGCGGACAG
GCCAAGTAGAACACGCGAAGCGC
TGGGCTGCCTGCTGCGACCAGGG
```



Spatio-Temporal Data

- Spatio-Temporal Data

Average Monthly Temperature of land and ocean



Types of Data by Structure

Structured data

- Tables, spreadsheets, databases
- Rows = objects, columns = attributes

Semi-structured data

- JSON, XML, logs

Unstructured data

- Text, images, audio, video

Examples:

- Student grades (structured)
- Web logs (semi-structured)
- Tweets, MRI images (unstructured)

Structured Data: The Tidy Spreadsheet

Data that adheres to a pre-defined data model. It's organized in a tabular format (rows and columns) where the attributes are clearly defined.

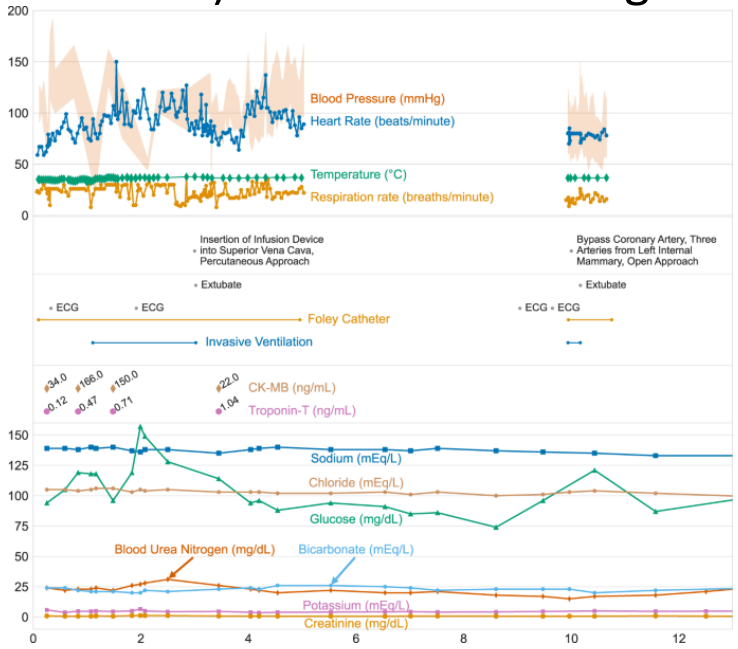
- **Record data** is the classic example of structured data.
- Easy for machines to process and analyze.
- Found in **relational databases** and spreadsheets.

Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
1	5.1	3.5	1.4	0.2	Iris-setosa
2	4.9	3	1.4	0.2	Iris-setosa
3	4.7	3.2	1.3	0.2	Iris-setosa
4	4.6	3.1	1.5	0.2	Iris-setosa
5	5	3.6	1.4	0.2	Iris-setosa
6	5.4	3.9	1.7	0.4	Iris-setosa
7	4.6	3.4	1.4	0.3	Iris-setosa
8	5	3.4	1.5	0.2	Iris-setosa
9	4.4	2.9	1.4	0.2	Iris-setosa
10	4.9	3.1	1.5	0.1	Iris-setosa
11	5.4	3.7	1.5	0.2	Iris-setosa
12	4.8	3.4	1.6	0.2	Iris-setosa
13	4.8	3	1.4	0.1	Iris-setosa
14	4.3	3	1.1	0.1	Iris-setosa
15	5.8	4	1.2	0.2	Iris-setosa
16	5.7	4.4	1.5	0.4	Iris-setosa
17	5.4	3.9	1.3	0.4	Iris-setosa
18	5.1	3.5	1.4	0.3	Iris-setosa
19	5.7	3.8	1.7	0.3	Iris-setosa

Unstructured Data: The Wild West

Data with no pre-defined model or organization. The attributes are not easily broken down into a table.

- Represents **over 80%** of all data!
- *Examples*: The text of an email, an image of a cat, an audio recording.
- Requires advanced techniques (like Natural Language Processing or Computer Vision) to extract meaningful attributes.



MIMIC-IV dataset

Semi-Structured Data

Contains tags or markers to separate semantic elements, providing a hierarchy but more flexibility than a rigid table.

- **JSON and XML** are the key examples.
- Commonly used in web APIs and document databases.

JSON

```
{
  "student": {
    "name": "Alex Doe",
    "major": "Data Science",
    "gpa": 3.9
  }
}
```

Student Info

json

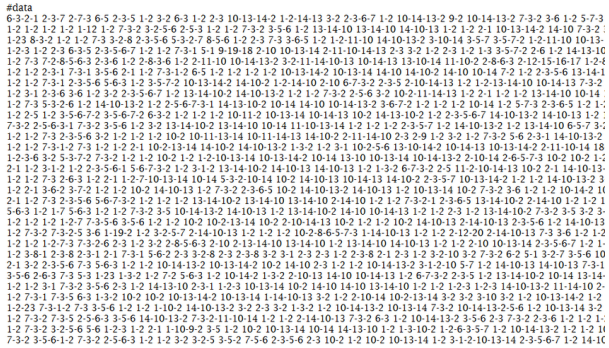
```
{
  "patient_id": "P1001",
  "name": {
    "first": "Alice",
    "last": "Wang"
  },
  "age": 45,
  "conditions": [
    {
      "name": "Hypertension",
      "diagnosed": "2020-05-12",
      "medications": [
        {"name": "Lisinopril", "dose": "10mg", "frequency": "daily"}
      ]
    },
    {
      "name": "Type 2 Diabetes",
      "diagnosed": "2018-11-03",
      "medications": [
        {"name": "Metformin", "dose": "500mg", "frequency": "twice daily"},
        {"name": "Insulin", "dose": "20 units", "frequency": "nightly"}
      ]
    }
  ],
  "allergies": ["Peanuts", "Penicillin"],
  "emergency_contact": {
    "name": "Bob Wang",
    "relationship": "Spouse",
    "phone": "+1-555-678-1234"
  }
}
```

Patient Health Records

Datasets: Collections of Data

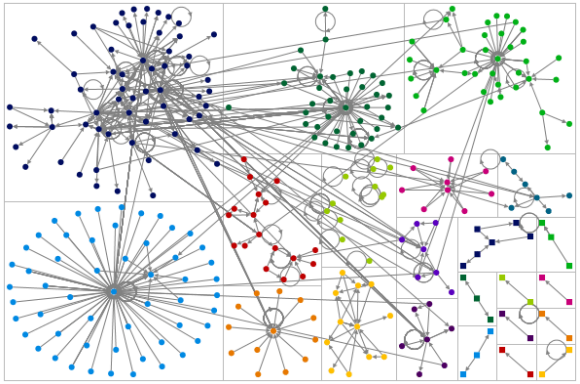
- A dataset is a structured collection of data, typically representing a set of data objects and their attributes.
- How we organize a dataset depends on the nature of the data and the relationships within it.





Taxes and Home Prices
http://lib.stat.cmu.edu/DASL/Stories/hometax.html

House	Sale price (100$)	Size (sqft)	Age (years)
Avalon	2050	2650	13
Cross Winds	2080	2600	*
The White House	2150	2554	6
The Rectory	2150	2921	3
Larchwood	1999	2580	4
Orchard House	1900	2580	4
Shangri-La	1800	2774	2
The Stables	1560	1920	1
Cobweb Cottage	1450	2150	*
Nairn House	1449	1710	1





Quick Question: Identify Data Types

For each example, identify the **data type**:

- Daily stock prices (Time Series)
- Shopping basket records (Transaction)
- GPS coordinates of Uber trips (Spatial)
- Email messages (Text)



Data Quality

Just because you have data doesn't mean you have *good* data. In the real world, data is often messy, incomplete, and inaccurate.

GIGO Principle: Garbage In, Garbage Out. 

If you feed a brilliant machine learning model low-quality data, you will get a useless, low-quality result.

Before we even think about analysis, we must ask: Can we trust our data?

Data Quality: The Good vs. The Bad

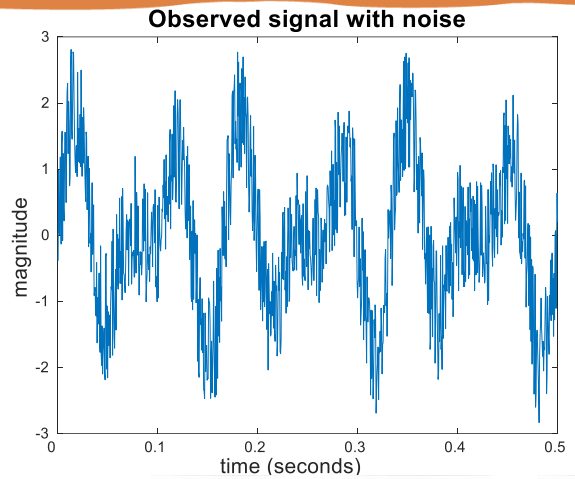
High-quality data is:

- Accurate
- Complete
- Consistent
- Timely
- Relevant

Low-quality data leads to:

- Biased models
- Incorrect decisions

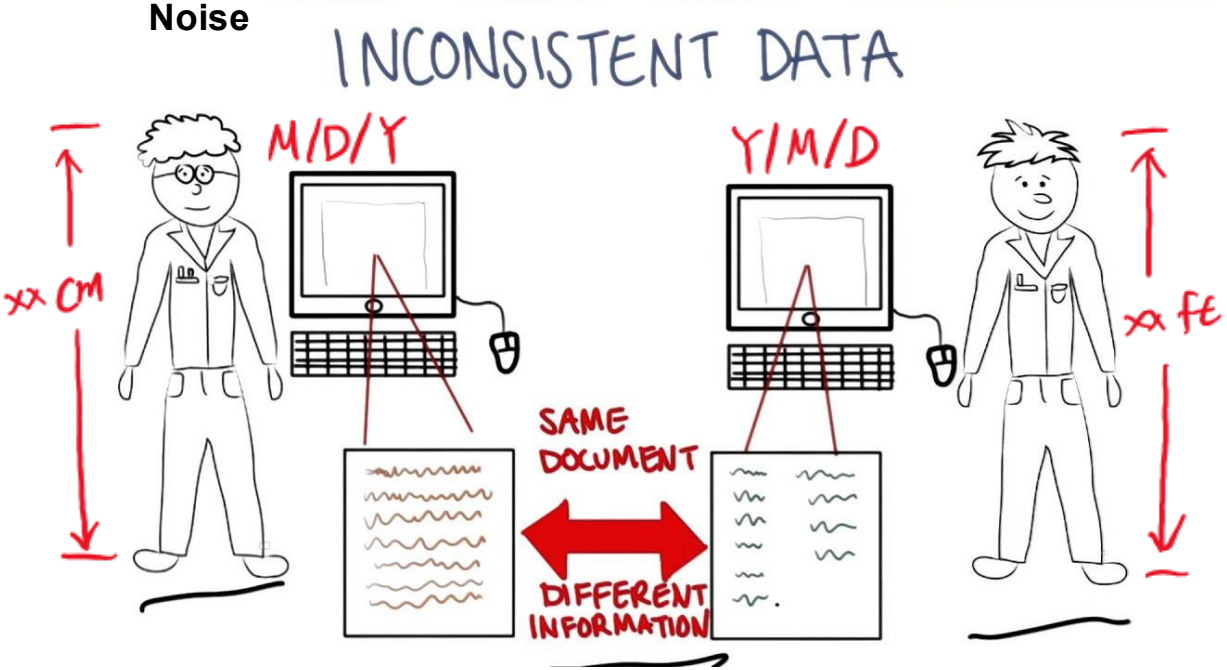
Common Data Quality Issues

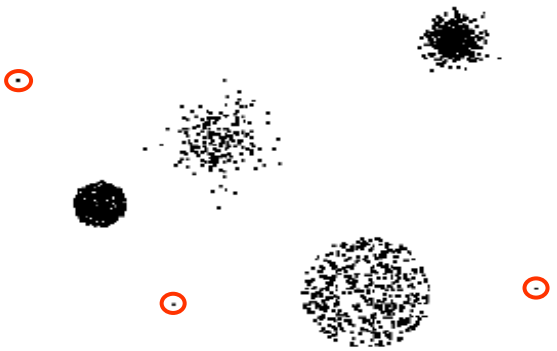


Noise

PERFORMANCE INDICATOR	YEAR	ANGLE	APUA	AQUALE	BEC	BEL	BELC	BLPC	BVE	CUC	DOMLE	EDF	ELMA	GBPC	GEBE	GRENLE
General Indicators																
Technical																
Service Coverage (%)	2002	99.8%	100.0%	100.0%		69.6%		100.0%		100.0%	96.0%		99.9%	100.0%	93.0%	98.0%
	2004		100.0%	100.0%		65.3%		100.0%			n.a.				93.0%	100.0%
	2005	97.5%	100.0%	100.0%		63.3%		100.0%	99.8%	100.0%	98.0%		100.0%		93.0%	
	2006	99.3%	100.0%	100.0%	100.0%	72.4%	100.0%	100.0%	99.6%	100.0%	100.0%	100.0%		100.0%	93.0%	98.9%
System Energy Losses (%)	2002	9.5%	18.5%	14.4%		11.5%		7.7%		4.4%	17.7%		4.5%	7.3%	9.0%	13.2%
	2004		26.6%	14.4%		12.7%		7.3%			14.6%				n.a.	9.7%
	2005	11.7%	33.1%	15.4%		13.4%		7.2%	9.4%	6.1%	18.0%		2.0%		7.9%	
	2006	9.9%	n.a.	15.0%	8.0%	13.9%	10.9%	7.5%	9.5%	n.a.	15.5%	38.0%		13.5%	8.1%	n.a.
System Load Factor (%)	2002	66.4%	64.8%	75.7%		67.0%		70.1%		66.5%	69.2%		80.0%	64.2%	75.7%	75.3%
	2004		58.7%			70.6%		69.3%			67.3%				n.a.	60.2%
	2005	76.3%	118.4%	74.5%		72.6%		66.4%	66.6%	65.2%	65.7%		77.6%		76.2%	
	2006	66.6%	122.0%	73.4%	101.3%	71.6%	69.1%	71.0%	73.5%	n.a.	65.4%	124.5%		67.7%	75.4%	n.a.

Missing Value





Outliers

The Dimensions of Data Quality

How do we measure data quality? We look at several key dimensions:

- **Accuracy**: Is the data correct and true to fact?
  - *Example: A person's age is listed as 150 years. This is likely inaccurate.*
- **Completeness**: Are there missing values?
  - *Example: 30% of customer records are missing an email address.*
- **Consistency**: Is the same data element consistent across different systems?
  - *Example: Does a patient's date of birth match in the hospital's admission and billing records?*
- **Timeliness**: Is the data recent enough for the analysis?
  - *Example: Using 10-year-old sales data to predict next month's trends is not timely.*
- **Noise level:** low measurement error
- **Bias:** representative of the population

Noise - (1)

- Noise refers to unwanted behaviors within the data that provide a low signal-to-noise ratio.
- Essentially, **data = signal + noise**.
- Noisy data are data is corrupted, distorted, or has a low signal-to-noise ratio.

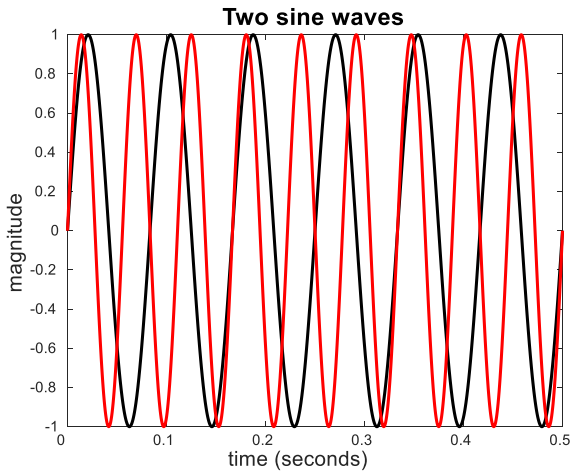
Signal-to-noise ratio is defined as the ratio of the power of a signal (meaningful input) to the power of background noise

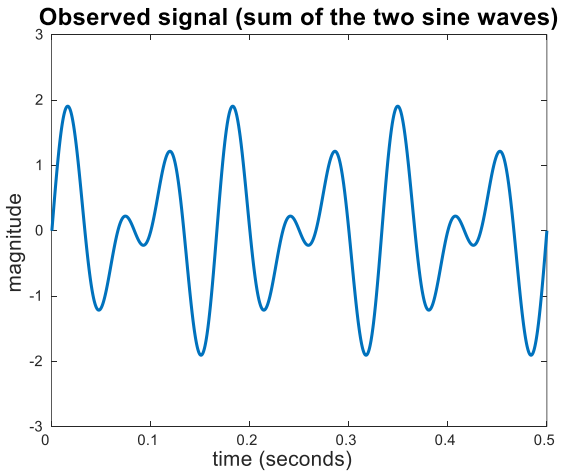


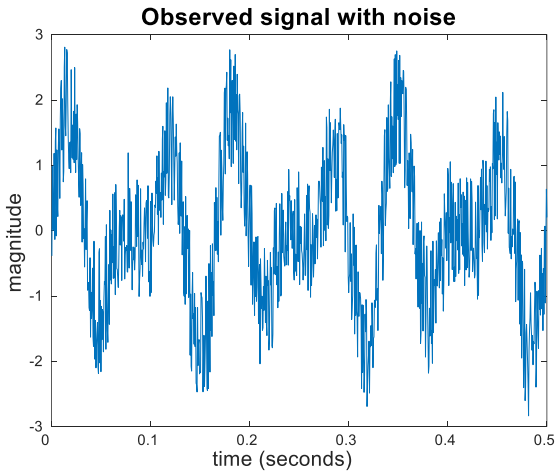


Noise - (2)

- For attributes, noise refers to modification of original values
  - Examples: distortion of a person's voice when talking on a poor phone and "snow" on television screen





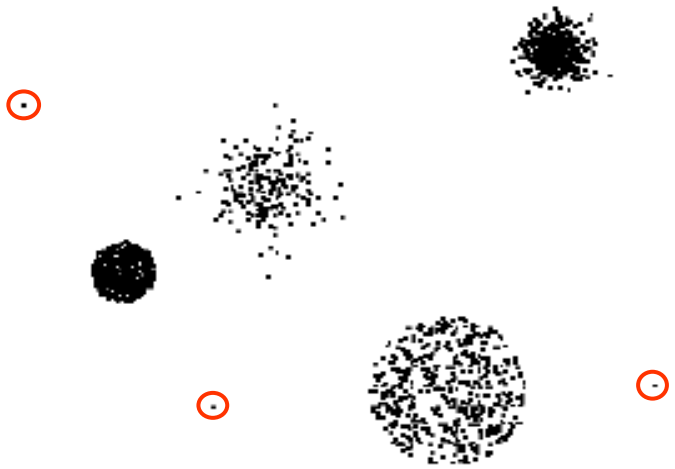


The figures below show two sine waves of the same magnitude and different frequencies, the waves combined, and the two sine waves with random noise

- The magnitude and shape of the original signal is distorted

Outliers: Case-1

- ***Outliers*** are data objects with characteristics that are considerably different than most of the other data objects in the data set
  - **Case 1:** Outliers are noise that interferes with data analysis



Outliers: Case-2

- ***Outliers*** are data objects with characteristics that are considerably different than most of the other data objects in the data set
  - **Case 2:** Outliers are the goal of our analysis
    - Credit card fraud
    - Intrusion detection

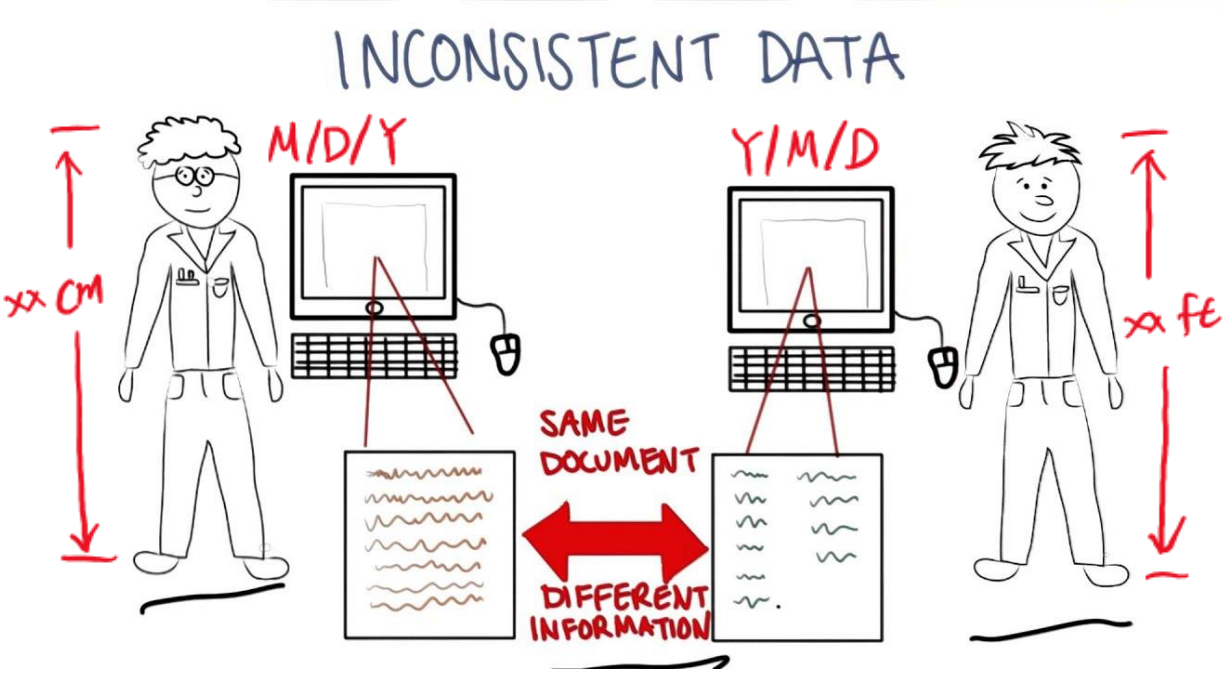


Credit card fraud has remained the most common type of identity theft through the first three quarters of 2023, with 318,000 reported cases.

Type of identity fraud	Reported cases, 2022	Percent change in reported cases, 2021 to 2022
Credit Card Fraud	440,666	13%
Other Identity Theft	326,505	(13%)
Bank Fraud	156,134	25%
Loan or Lease Fraud	153,578	(22%)
Employment or Tax-Related Fraud	103,416	(7%)
Phone or Utilities Fraud	77,316	(13%)
Government Documents or Benefits Fraud	57,912	(85%)

Inconsistent Data

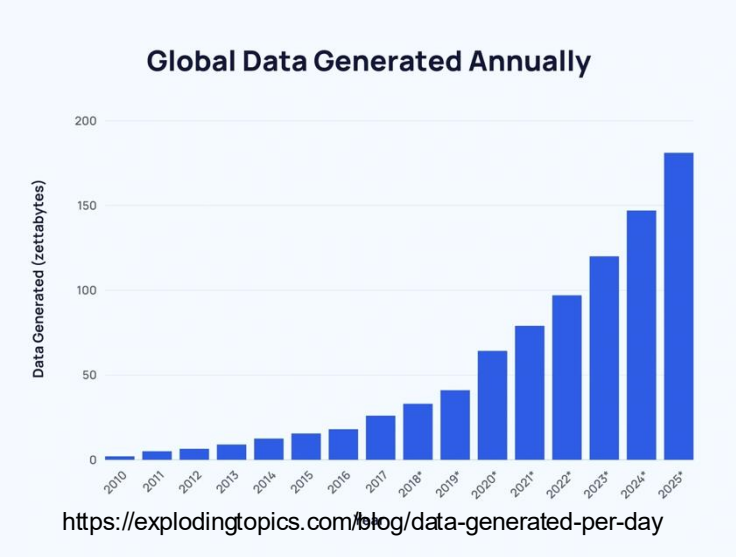
- When work with multiple data sources, it's likely to have mismatches in the same information across sources.
- Inconsistent data can also get introduced during migration or company mergers, etc.
- If not reconciled constantly, inconsistencies in data tend to destroy the value of data.



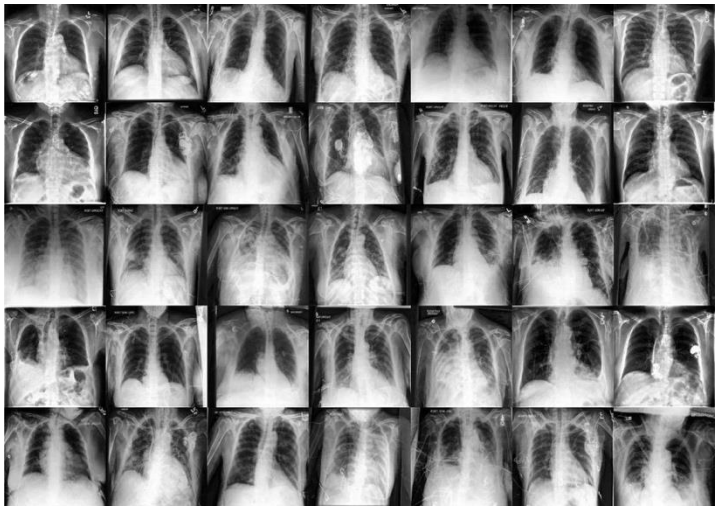
Generated Data: The Good

Generated data (synthetic data) is the data that is not actually from the real events but is generated by computer software yet emulates the mathematical and statistical characteristics –alongside the structure and format- of real-world datasets.

- Pros:
  - Efficiency: generating and using generated data is a lot easier than using real data. Generating data is a cost-effective way to collect a massive data input for training AI/ML models;
  - Protect the privacy of real data;



https://explodingtopics.com/blog/data-generated-per-day



Generated Data: The Issues

Generated data (synthetic data) is the data that is not actually from the real events but is generated by computer software yet emulates the mathematical and statistical characteristics –alongside the structure and format- of real-world datasets.

- Cons:
  - It is often difficult to statistically validate whether synthetic dataset is true to the structure and format of the dataset it emulates.
  - Generated fake data can be misused (fake video, image, misinformation, etc)





Missing Values

- Different types for missing values (unknown, unrecorded, irrelevant), i.e.,
  - Information is not collected (e.g., people decline to give their age and weight)
  - Attributes may not be applicable to all cases (e.g., annual income is not applicable to children)
- Reasons:
  - Malfunctioning equipment
  - Changes in experimental design
  - Collation of different datasets
  - Measurement not possible.

PERFORMANCE INDICATOR	YEAR	ANGLE	APUA	AQUALEC	BEC	BEL	BELCC	BLPC	BVIE	CUC	DOMLE	EOF	ELMAI	GBPC	GEBE	GRENLE
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	2005	11.7%	33.1%	15.4%		13.4%		7.2%	9.4%	6.1%	18.0%		2.0%		7.9%	
	2006	9.9%	n.a.	15.0%	8.0%	13.9%	10.9%	7.5%	9.5%	n.a.	15.5%	36.0%		13.5%	8.1%	n.a.
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	2004		58.7%			70.6%		69.3%			67.3%				n.a.	60.2%
	2005	70.3%	116.4%	74.5%		72.6%		66.4%	66.6%	65.2%	65.7%		77.8%		76.2%	
	2006	68.6%	122.0%	73.4%	101.3%	71.6%	69.1%	71.0%	73.5%	n.a.	65.4%	124.5%		67.7%	75.4%	n.a.

Duplicate Data

- Data set may include data objects that are duplicates, or almost duplicates of one another
  - Major issue when merging data from heterogeneous sources

- Examples:
  - Same person with multiple email addresses

Data Quality: What can we do about these problems?

- Filter out noise
- Detect and remove outliers
- Define rules to merger data if collected through different parties;
- Handling missing values
  - Eliminate data objects or variables
  - Estimate missing values (mean, average, etc)
    - Example: time series of temperature
    - Example: census results
  - Ignore the missing value during analysis
- Remove duplicated data

Quick Question: Missing Data

Which strategy is **usually NOT recommended** when handling missing data?

A. Removing records with many missing values

B. Filling missing values with the mean

C. Filling missing values with random numbers

D. Using model-based imputation



Quick Question: Outlier

Scenario

A patient's heart rate is recorded as **250 bpm**.

Question:

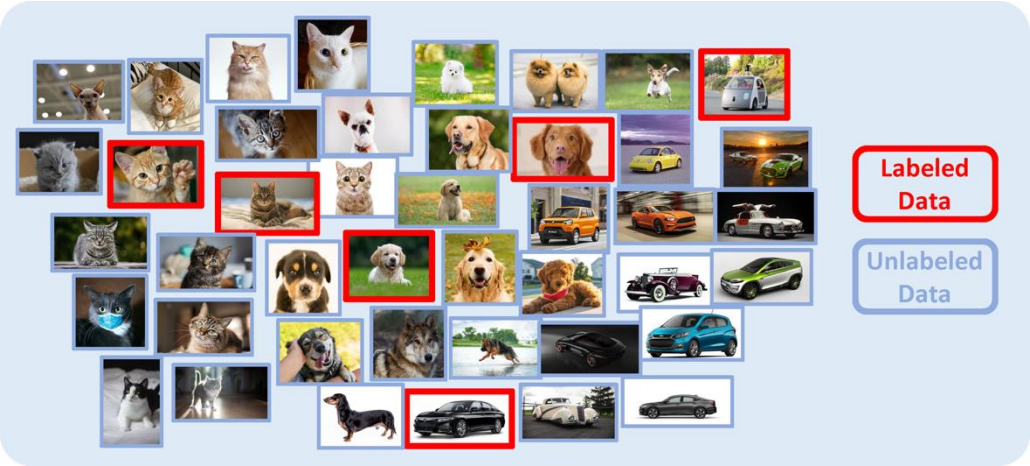
What should you do first?

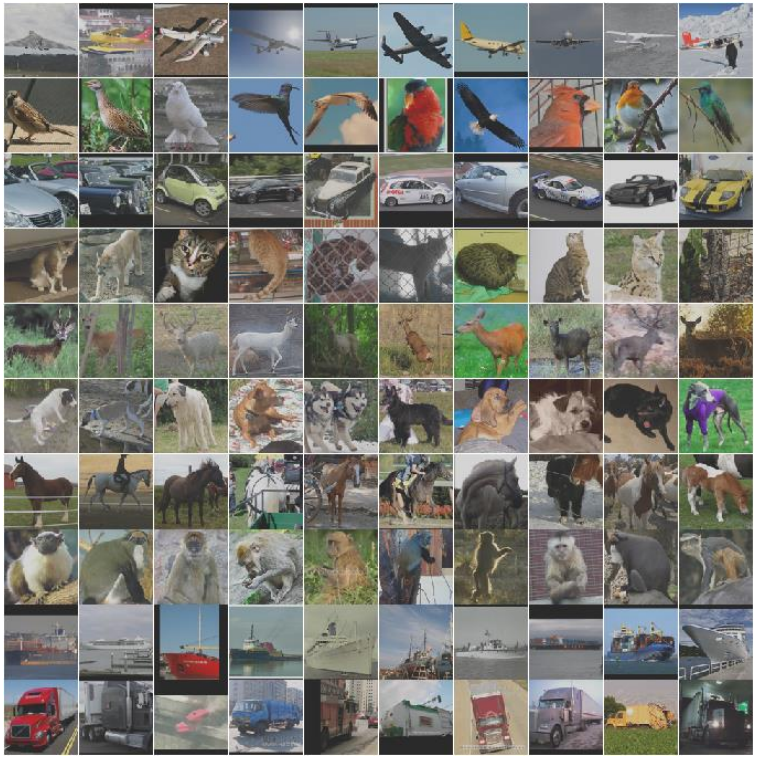
A. Remove it immediately
B. Keep it without question
C. Investigate whether it is an error or rare event
D. Replace it with the average



Labeled vs unlabeled data

- Data can be labeled or unlabeled (Classification)

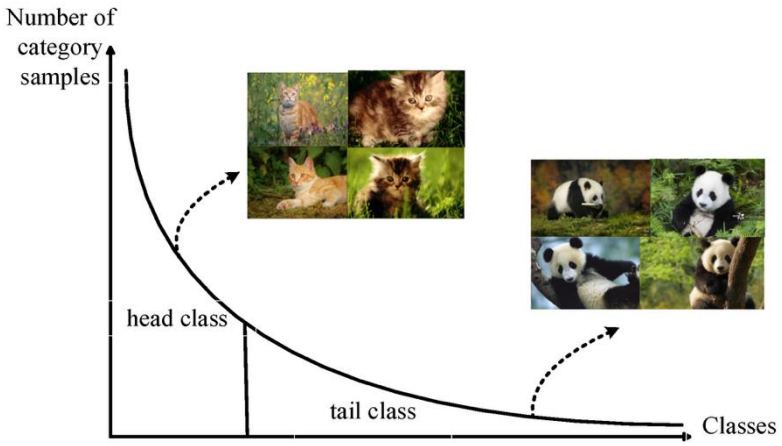


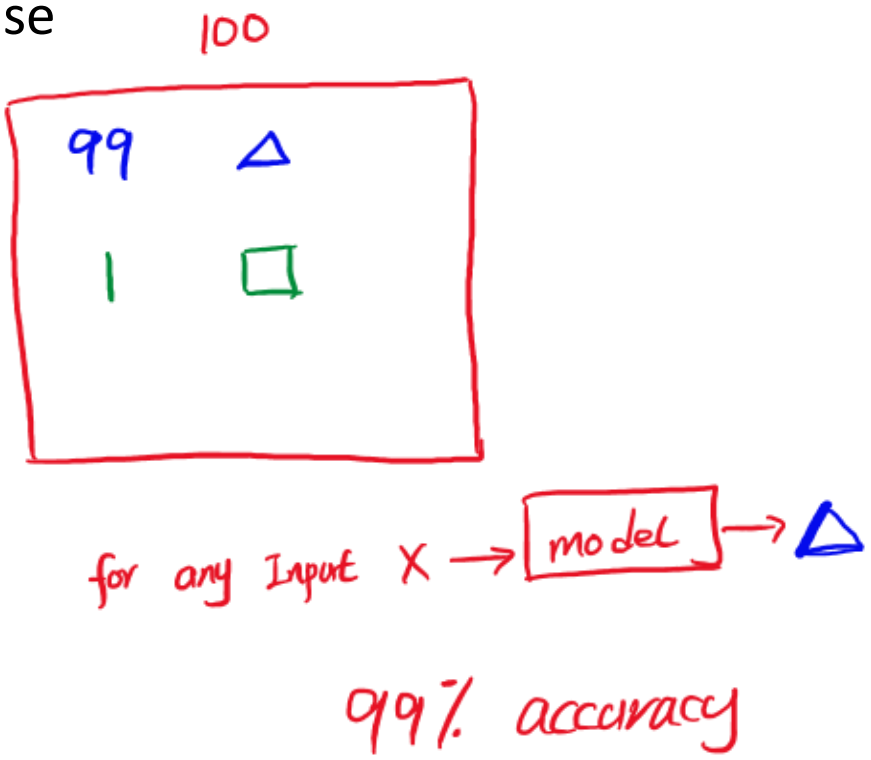


Data annotation (manually) can be extremely expensive;
Most of data are unlabeled;

Unbalanced data

- Unbalanced data is a well-known problem in classification problems
  - One class is often far more prevalent than the rest
  - Example: detecting a rare disease





Unbalanced data (Cont'd)

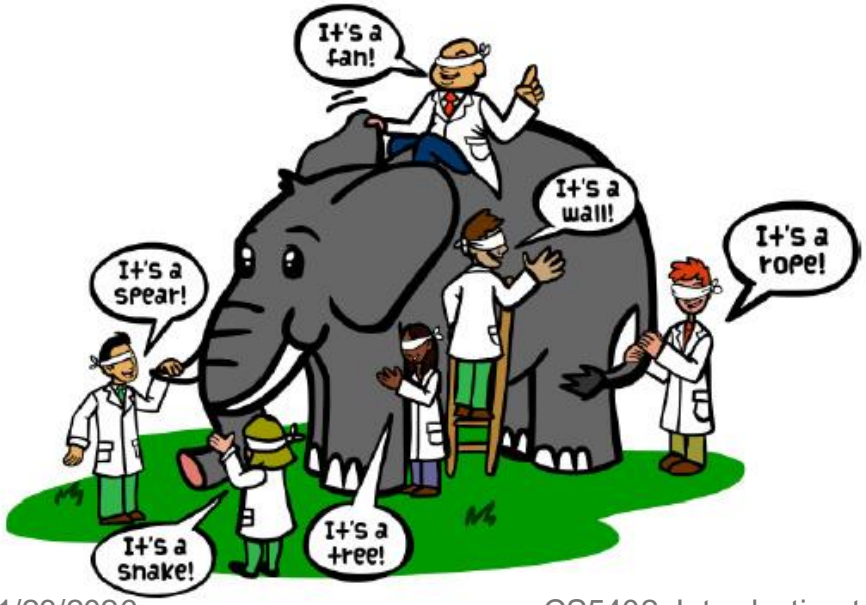
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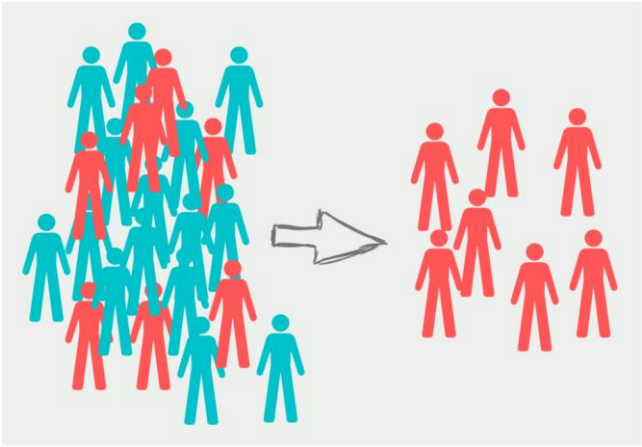
- Main problem: simply predicting the majority class yields high accuracy but is not useful
  - Predicting that no patient has the rare disease gives high classification accuracy
- Unbalanced data requires techniques that can deal with unequal misclassification costs
  - Misclassifying an afflicted patient may be much more costly than misclassifying a healthy one

Data bias

- Data bias occurs when data or information is limited in some way, painting an inaccurate representation of the population, or doesn't tell the full story.

"today's theories were derived from scant data and few, frequently occurring cases, after which they were modelled. It is thus no wonder that they fail to predict new occurrences." Described by Francis Bacon 400 years ago.





Getting to know your data.

- Simple visualization tools are very useful
  - Nominal attributes: histograms (Is the distribution consistent with background knowledge?)
  - Numeric attributes: graphs
    (Any obvious outliers?)
- 2-D and 3-D plots show dependencies
- May need to consult domain experts
- Too much data to inspect manually? Take a sample!

Data Quality vs. Model Complexity

- High-quality data + simple model
  - often better than
- Low-quality data + complex model

Summary

- Data types determine what methods you can use
- Attribute types affect distance, similarity, and models
- Data quality is critical for trustworthy results
- Real-world data is messy by default

Next Lecture:

- Data preprocessing
- Data normalization
- Feature selection